

# Отчёт по лабораторной работе №10

Дисциплина: Администрирование сетевых подсистем

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## Цель работы

Целью данной работы является приобретение практических навыков по конфигурированию SMTP-сервера в части настройки аутентификации.

## Выполнение лабораторной работы

### Подготовка к работе

На виртуальной машине server войдём под нашим пользователем и откроем терминал. Перейдём в режим суперпользователя (рис. @fig-1):

### Мониторинг почтовой службы

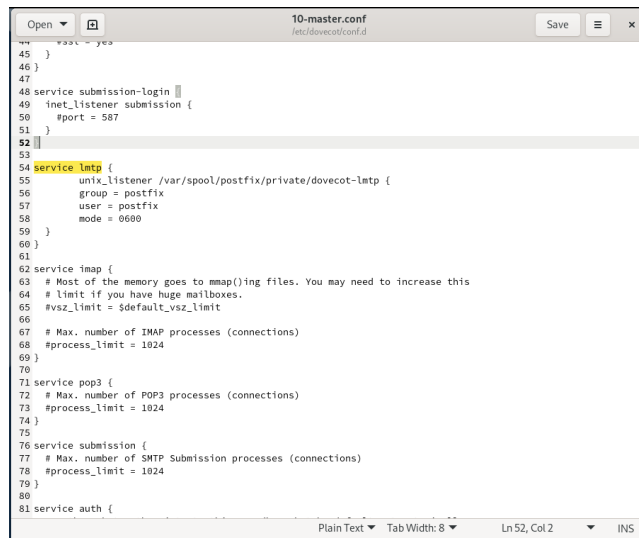
В дополнительном терминале запустим мониторинг работы почтовой службы (рис. @fig-2):

### Настройка LMTP в Dovecot

Добавим в список протоколов, с которыми может работать Dovecot, протокол LMTP (рис. @fig-3):

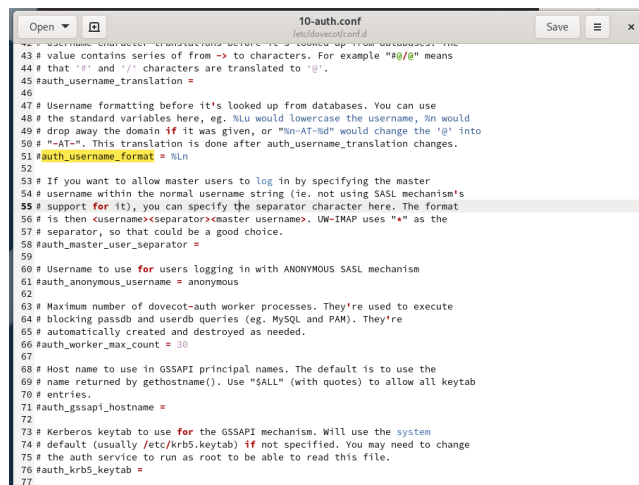
### Настройка сервиса lmtp

Настроим в Dovecot сервис lmtp для связи с Postfix, определив расположение unix-сокета и права доступа (рис. @fig-4):



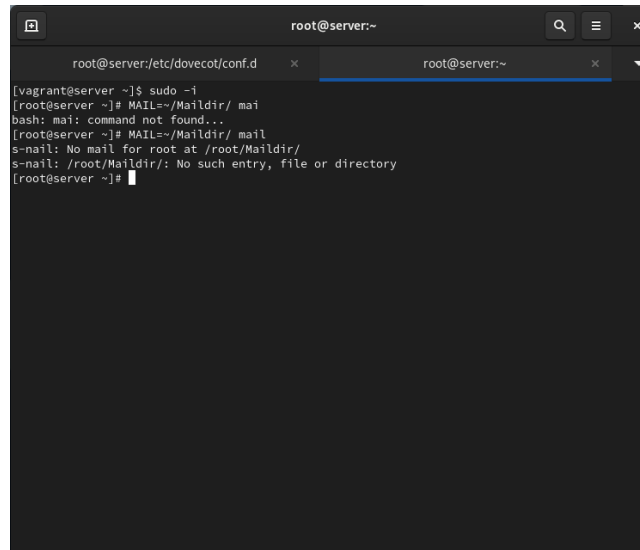
```
44 #ask = yes
45 }
46 }
47
48 service submission-login {
49   #inet_listener submission {
50     #port = 587
51   }
52 }
53
54 service lmtop {
55   unix_listener /var/spool/postfix/private/dovecot-lmtop {
56     group = postfix
57     user = postfix
58     mode = 0600
59   }
60 }
61
62 service imap {
63   # Host of the memory goes to mmap()ing files. You may need to increase this
64   # limit if you have huge mailboxes.
65   #vsz_limit = $default_vsz_limit
66
67   # Max. number of IMAP processes (connections)
68   #process_limit = 1024
69 }
70
71 service pop3 {
72   # Max. number of POP3 processes (connections)
73   #process_limit = 1024
74 }
75
76 service submission {
77   # Max. number of SMTP Submission processes (connections)
78   #process_limit = 1024
79 }
80
81 service auth {
```

Figure 1: Открытие режима суперпользователя на виртуальной машине server.



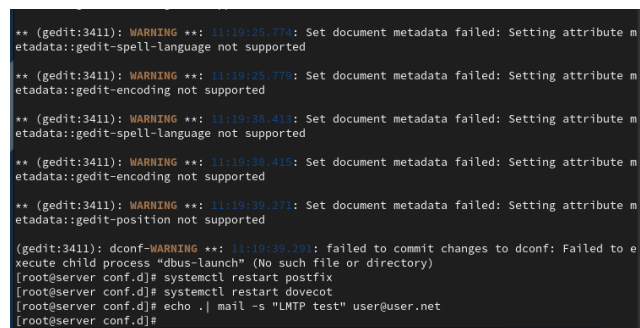
```
43 # value contains series of from -> to characters. For example "#@/@" means
44 # that '#' and '/' characters are translated to '@'.
45 #auth_username_translation =
46
47 # Username formatting before it's looked up from databases. You can use
48 # the standard variables here, eg. %lu would lowercase the username, %n would
49 # drop away the domain if it was given, or "%n-AT-%d" would change the '@' into
50 # "-AT-". This translation is done after auth_username_translation changes.
51 #auth_username_format = %Ln
52
53 # If you want to allow master users to log in by specifying the master
54 # username within the normal username string (ie. not using SASL mechanism's
55 # support for it), you can specify the separator character here. The format
56 # is then <username>separator<master username>. UN-IMAP uses "*" as the
57 # separator, so that could be a good choice.
58 #auth_master_user_separator =
59
60 # Username to use for users logging in with ANONYMOUS SASL mechanism
61 #auth_anonymous_username = anonymous
62
63 # Maximum number of dovecot-auth worker processes. They're used to execute
64 # blocking passdb and userdb queries (eg. MySQL and PAM). They're
65 # automatically created and destroyed as needed.
66 #auth_worker_max_count = 30
67
68 # Host name to use in GSSAPI principal names. The default is to use the
69 # name returned by gethostname(). Use "SALL" (with quotes) to allow all keytab
70 # entries.
71 #auth_gssapi_hostname =
72
73 # Kerberos keytab to use for the GSSAPI mechanism. Will use the system
74 # default (usually /etc/krb5.keytab) if not specified. You may need to change
75 # the auth service to run as root to be able to read this file.
76 #auth_krb5_keytab =
77
```

Figure 2: Запуск мониторинга работы почтовой службы.



```
root@server:~  
root@server:~/etc/dovecot/conf.d  
[vagrant@server ~]$ sudo -i  
[root@server ~]# MAIL=~/.Maildir/ mai  
bash: mai: command not found...  
[root@server ~]# MAIL=~/.Maildir/ mail  
s-nail: No mail for root at /root/.Maildir/  
s-nail: /root/.Maildir/: No such entry, file or directory  
[root@server ~]#
```

Figure 3: Добавление протокола LMTP в Dovecot.

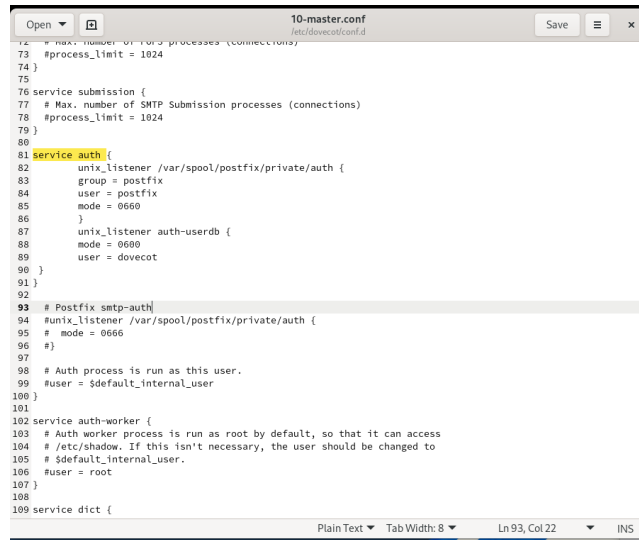


```
** (gedit:3411): WARNING **: 11:19:25.774: Set document metadata failed: Setting attribute m  
etadate::gedit-spell-language not supported  
** (gedit:3411): WARNING **: 11:19:25.779: Set document metadata failed: Setting attribute m  
etadate::gedit-encoding not supported  
** (gedit:3411): WARNING **: 11:19:38.412: Set document metadata failed: Setting attribute m  
etadate::gedit-spell-language not supported  
** (gedit:3411): WARNING **: 11:19:38.415: Set document metadata failed: Setting attribute m  
etadate::gedit-encoding not supported  
** (gedit:3411): WARNING **: 11:19:39.271: Set document metadata failed: Setting attribute m  
etadate::gedit-position not supported  
(gedit:3411): dconf-WARNING **: 11:19:39.391: failed to commit changes to dconf: Failed to e  
xecute child process "dbus-launch" (No such file or directory)  
[root@server conf.d]# systemctl restart postfix  
[root@server conf.d]# systemctl restart dovecot  
[root@server conf.d]# echo . | mail -s "LMTP test" user@user.net  
[root@server conf.d]#
```

Figure 4: Настройка сервиса lmtp в Dovecot.

## Настройка Postfix для работы с LMTP

Переопределим в Postfix передачу сообщений через заданный unix-сокет (рис. @fig-5):



```
73 # Max. number of LMTP processes (connections)
74 #process_limit = 1024
75 }
76 service submission {
77 # Max. number of SMTP Submission processes (connections)
78 #process_limit = 1024
79 }
80
81 service auth {
82   unix_listener /var/spool/postfix/private/auth {
83     group = postfix
84     user = postfix
85     mode = 0660
86   }
87   unix_listener auth-userdb {
88     mode = 0660
89     user = dovecot
90   }
91 }
92
93 # Postfix smtp-auth
94 #unix_listener /var/spool/postfix/private/auth {
95 #  mode = 0660
96 #}
97
98 # Auth process is run as this user.
99 #user = $default_internal_user
100 }
101
102 service auth-worker {
103 # Auth worker process is run as root by default, so that it can access
104 # /etc/shadow. If this isn't necessary, the user should be changed to
105 # $default_internal_user.
106 #user = root
107 }
108
109 service dict {
```

Figure 5: Настройка Postfix для работы с LMTP.

## Настройка формата аутентификации

В файле /etc/dovecot/conf.d/10-auth.conf зададим формат имени пользователя для аутентификации в форме логина без указания домена (рис. @fig-6):

## Перезапуск служб

Перезапустим Postfix и Dovecot для применения изменений (рис. @fig-7):

## Отправка тестового письма

Из-под учётной записи своего пользователя отправим письмо с клиента (рис. @fig-8):

## Просмотр логов и почтового ящика

Посмотрим содержание логов при мониторинге почтовой службы и проверим почтовый ящик пользователя на сервере

```
root@server:/etc/dovecot/conf.d

** (gedit:3650): WARNING **: 11:30:26.880: Set document metadata failed: Setting attribute
etadate::gedit-encoding not supported

** (gedit:3650): WARNING **: 11:30:26.288: Set document metadata failed: Setting attribute
etadate::gedit-spell-language not supported

** (gedit:3650): WARNING **: 11:30:26.290: Set document metadata failed: Setting attribute
etadate::gedit-encoding not supported

** (gedit:3650): WARNING **: 11:30:30.857: Set document metadata failed: Setting attribute
etadate::gedit-spell-language not supported

** (gedit:3650): WARNING **: 11:30:30.861: Set document metadata failed: Setting attribute
etadate::gedit-encoding not supported

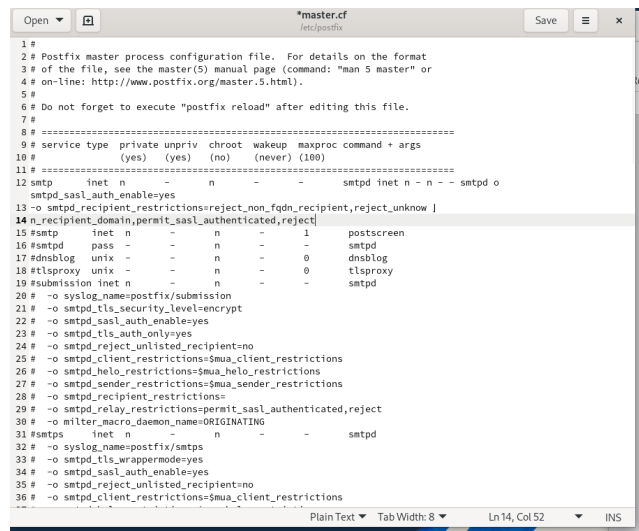
** (gedit:3650): WARNING **: 11:30:30.867: Set document metadata failed: Setting attribute
etadate::gedit-position not supported

(gedit:3650): dconf-WARNING **: 11:30:30.941: failed to commit changes to dconf: Failed to
xecute child process "dbus-launch" (No such file or directory)
[root@server conf.d]# postconf -e 'smtpd_sasl_type = dovecot'
[root@server conf.d]# postconf -e 'smtpd_sasl_path = private/auth'
[root@server conf.d]# postconf -e 'smtpd_recipient_restrictions =
reject_unknown_recipient_domain, permit_mynetworks,
reject_non_fqdn_recipient, reject_unauth_destination,
reject_unverified_recipient, permit'
postconf: fatal: -e, -X, or -# accepts no multi-line input
[root@server conf.d]# postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient
omain, permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination, reject_un
verified_recipient, permit'
[root@server conf.d]# postconf -e 'mynetworks = 127.0.0.0/8'
[root@server conf.d]#
```

Figure 6: Настройка формата аутентификации в Dovecot.

```
1 #
2 # Postfix master process configuration file. For details on the format
3 # of the file, see the master(5) manual page (command: "man 5 master" or
4 # on-line: http://www.postfix.org/master.5.html).
5 #
6 # Do not forget to execute "postfix reload" after editing this file.
7 #
8 # =====
9 # service type private unpriv chroot wakeup maxproc command + args
10 # (yes) (yes) (no) (never) (100)
11 # =====
12 smtp inet n - n - - smtpd
13 #smtp inet n - n - 1 postscreen
14 #smtpd pass - - n - - smtpd
15 #dnsmlog unix - - n - 0 dnsmlog
16 #tlspd unix - - n - 0 tlspd
17 #submission inet n - n - - smtpd
18 # -o syslog_name=postfix/submission
19 # -o smtpd_tls_security_level=encrypt
20 # -o smtpd_sasl_auth_enable=yes
21 # -o smtpd_tls_auth_only=yes
22 # -o smtpd_reject_unlisted_recipient=no
23 # -o smtpd_client_restrictions=$mua_client_restrictions
24 # -o smtpd_helo_restrictions=$mua_helo_restrictions
25 # -o smtpd_sender_restrictions=$mua_sender_restrictions
26 # -o smtpd_recipient_restrictions=
27 # -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
28 # -o milter_macro_daemon_name=ORIGINATING
29 #smtps inet n - n - - smtpd
30 # -o syslog_name=postfix/smtps
31 # -o smtpd_tls_wrappermode=yes
32 # -o smtpd_sasl_auth_enable=yes
33 # -o smtpd_reject_unlisted_recipient=no
34 # -o smtpd_client_restrictions=$mua_client_restrictions
35 # -o smtpd_helo_restrictions=$mua_helo_restrictions
36 # -o smtpd_sender_restrictions=$mua_sender_restrictions
37 # -o smtpd_recipient_restrictions=
```

Figure 7: Перезапуск Postfix и Dovecot.



```
1 #
2 # Postfix master process configuration file. For details on the format
3 # of the file, see the master(5) manual page (command: "man 5 master" or
4 # on-line: http://www.postfix.org/master.5.html).
5 #
6 # Do not forget to execute "postfix reload" after editing this file.
7 #
8 # =====
9 # service type private unpriv chroot wakeup maxproc command + args
10 # (yes) (yes) (no) (never) (100)
11 # =====
12 smtp inet n - n - - smtpd inet n - n - - smtpd o
13 smtpd_sasl_auth_enable=yes
14 n_recipient_restrictions=reject_non_fqdn_recipient,reject_unknown ]
15 #smtpd inet n - n - - 1 postscreen
16 #smtpd pass - - n - - smtpd
17 #dnsblog unix - - n - 0 dnsblog
18 #tlspolicy unix - - n - 0 tlspolicy
19 #submission inet n - n - - smtpd
20 # -o syslog_name=postfix/submission
21 # -o smtpd_tls_security_level=encrypt
22 # -o smtpd_sasl_auth_enable=yes
23 # -o smtpd_tls_auth_only=yes
24 # -o smtpd_reject_unlisted_recipient=no
25 # -o smtpd_client_restrictions=$mua_client_restrictions
26 # -o smtpd_helo_restrictions=$mua_helo_restrictions
27 # -o smtpd_sender_restrictions=$mua_sender_restrictions
28 # -o smtpd_recipient_restrictions=
29 # -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
30 # -o milter_macro_daemon_name=ORIGINATING
31 #smtps inet n - n - - smtpd
32 # -o syslog_name=postfix/smtps
33 # -o smtpd_tls_wrappermode=yes
34 # -o smtpd_sasl_auth_enable=yes
35 # -o smtpd_reject_unlisted_recipient=no
36 # -o smtpd_client_restrictions=$mua_client_restrictions
```

Figure 8: Отправка тестового письма с клиента.

(рис. @fig-9, @fig-10):

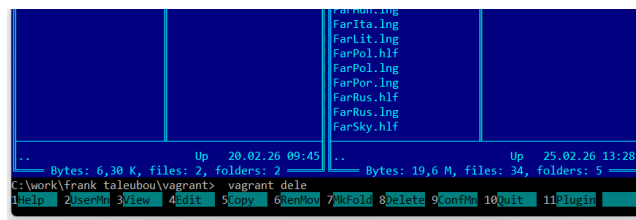


Figure 9: Просмотр логов почтовой службы.

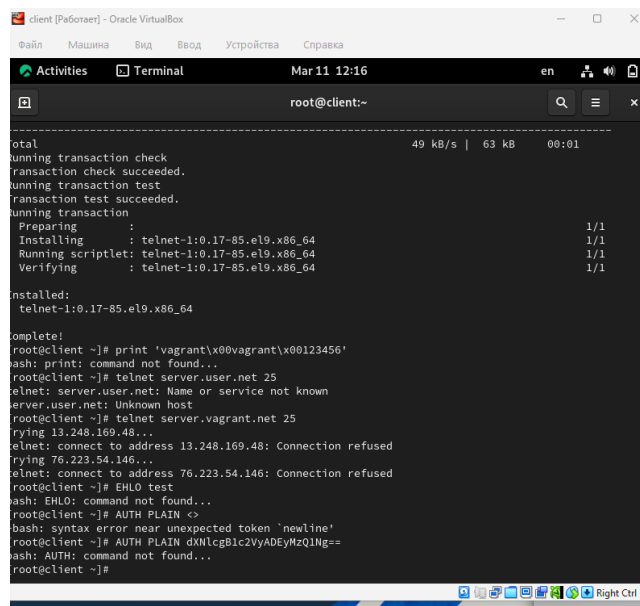
## Настройка аутентификации SASL

Определим службу аутентификации пользователей в Dovecot и настроим Postfix для использования SASL-аутентификации (рис. @fig-11):

## Временный запуск SMTP с аутентификацией

Для проверки работы аутентификации временно запустим SMTP-сервер с возможностью аутентификации (рис. @fig-12):





```
total 49 kB/s | 63 kB 00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : telnet-1:0.17-85.el9.x86_64 1/1
Running scriptlet: telnet-1:0.17-85.el9.x86_64 1/1
Verifying : telnet-1:0.17-85.el9.x86_64 1/1

Installed:
telnet-1:0.17-85.el9.x86_64

complete!
root@client ~# print 'vagrant\x00vagrant\x00123456'
bash: print: command not found...
root@client ~# telnet server.user.net 25
telnet: server.user.net: Name or service not known
server.user.net: Unknown host
root@client ~# telnet server.vagrant.net 25
Trying 13.248.169.48...
telnet: connect to address 13.248.169.48: Connection refused
Trying 76.223.54.146...
telnet: connect to address 76.223.54.146: Connection refused
root@client ~# EHLO test
bash: EHLO: command not found...
root@client ~# AUTH PLAIN <>
bash: syntax error near unexpected token 'newline'
root@client ~# AUTH PLAIN dXNlcg8lc2VyADEyMzQlNg==
bash: AUTH: command not found...
root@client ~#
```

Figure 12: Настройка SMTP-сервера для проверки аутентификации.

## Тестирование аутентификации через telnet

На клиенте установим telnet, получим строку для аутентификации и протестируем подключение к SMTP-серверу (рис. @fig-13):

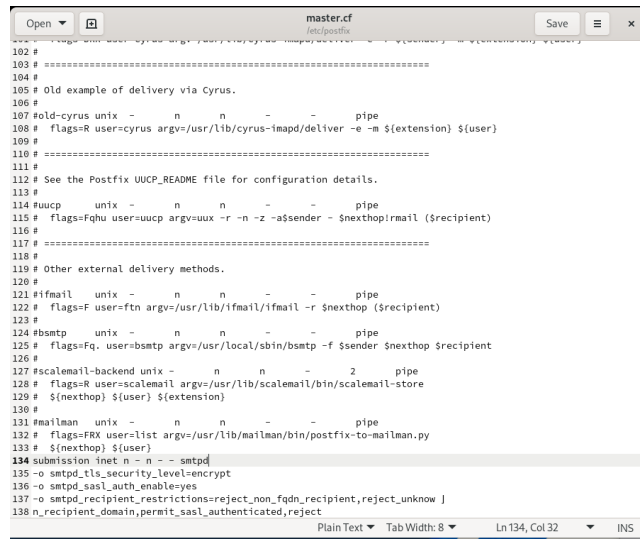
## Настройка SMTP на порту 587

Настроим SMTP-сервер на 587-м порту, разрешим соответствующую службу в межсетевом экране и перезапустим Postfix (рис. @fig-14):

## Выводы

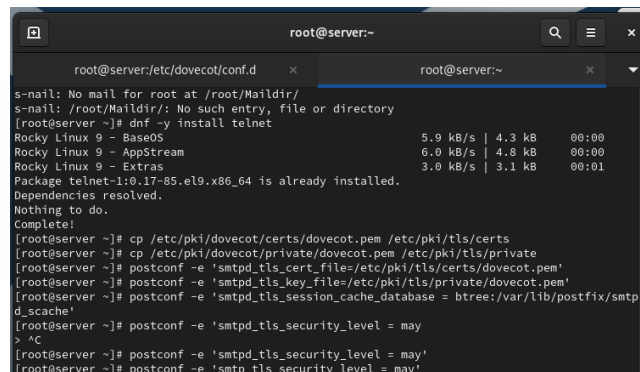
В ходе выполнения лабораторной работы были приобретены практические навыки по конфигурированию SMTP-сервера в части настройки аутентификации.





```
102 #
103 #
104 #
105 # Old example of delivery via Cyrus.
106 #
107 #old-cyrus unix - n n - - pipe
108 # flags=R user=cyrus argv=/usr/lib/cyrus-imapd/deliver -e -m ${extension} ${user}
109 #
110 #
111 #
112 # See the Postfix UUCP_README file for configuration details.
113 #
114 #uucp unix - n n - - pipe
115 # flags=Fqhu user=uucp argv=uux -r -n -z -a$sender - $nexthop!rmail ($recipient)
116 #
117 #
118 #
119 # Other external delivery methods.
120 #
121 #ifmail unix - n n - - pipe
122 # flags=F user=ftn argv=/usr/lib/ifmail/ifmail -r $nexthop ($recipient)
123 #
124 #bsmtp unix - n n - - pipe
125 # flags=Fq. user=bsmtp argv=/usr/local/sbin/bsmtp -f $sender $nexthop $recipient
126 #
127 #scalemail-backend unix - n n - 2 pipe
128 # flags=R user=scalemail argv=/usr/lib/scalemail/bin/scalemail-store
129 # $(nexthop) $(user) ${extension}
130 #
131 #mailman unix - n n - - pipe
132 # flags=FRX user=list argv=/usr/lib/mailman/bin/postfix-to-mailman.py
133 # $(nexthop) $(user)
134 submission inet n - n - - smtpd
135 -o smtpd_tls_security_level=encrypt
136 -o smtpd_sasl_auth_enable=yes
137 -o smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknown_
138 n_recipient_domain,permit_sasl_authenticated,reject
```

Figure 13: Тестирование аутентификации через telnet.



```
root@server:~
root@server:~/etc/dovecot/conf.d
s-mail: No mail for root at /root/Maildir/
s-mail: /root/Maildir/: No such entry, file or directory
[root@server ~]# dnf -y install telnet
Rocky Linux 9 - BaseOS                    5.9 kB/s | 4.3 kB    00:00
Rocky Linux 9 - AppStream                  6.0 kB/s | 4.8 kB    00:00
Rocky Linux 9 - Extras                     3.0 kB/s | 3.1 kB    00:01
Package telnet-1:0.17-85.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server ~]# cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs
[root@server ~]# cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private
[root@server ~]# postconf -e 'smtpd_tls_cert_file=/etc/pki/tls/certs/dovecot.pem'
[root@server ~]# postconf -e 'smtpd_tls_key_file=/etc/pki/tls/private/dovecot.pem'
[root@server ~]# postconf -e 'smtpd_tls_session_cache_database = btree:/var/lib/postfix/smtpd_scache'
[root@server ~]# postconf -e 'smtpd_tls_security_level = may'
> ^C
[root@server ~]# postconf -e 'smtpd_tls_security_level = may'
[root@server ~]# postconf -e 'smtpd_tls_security_level = may'
```

Figure 14: Настройка SMTP на порту 587.

## Контрольные вопросы

1. **Приведите пример задания формата аутентификации пользователя в Dovecot в форме логина с указанием домена.**

В конфигурационном файле Dovecot (/etc/dovecot/conf.d/10-auth.conf) можно указать формат аутентификации следующим образом: